May-26

Day-41

Syslog

- Syslog is a protocol used to keep logs of various events that happen on the device
- Log msgs can be shown in real time in the CLI of the device to inform us about important events as they occur.

Syslog.

- Syslog is an industry standard protocol for message logging.
- On network devices, Syslog can be used to log events such as changes in interface status (up↔ down), changes in OSPF neighbor status (up↔ down), system restarts etc.
- The msgs can be displayed in CLI as we configure the device, saved in RAM, or sent to an external Syslog server.

Eg:
```
int g0/0
no shutdown
```
Sample: `*.LINK-3-UPDOWN: ----` → this is syslog msg in CLI.

- Logs are essential when troubleshooting issues, examining the cause of incidents, etc.
- Syslog and SNMP are both used for monitoring and troubleshooting of devices. They are complementary, but their functionalities are different.

Syslog Message format

Seq: time stamp: %-facility-severity-MNEMONIC: description

- Seq: Seq of no. indicating the order/sequence of message
- time stamp: Time stamp indicating when msg was generated

- seqno: a seq no indicating the order/sequence of mssgs.
- Timestamp - These timestamps are particularly important especially when comparing logs of different devices. Assuming both devices have accurate time, we can use timestamps to compare when diffrnt events occured on diffrnt devices.

* NOB: seq & timestamps may or mayn't be displayed depending on device's configuration. Seq are generally not used, but its recommended to use timestamp as they are very important.

Facility: a value that indicates which process on the device gererated the message. Eg: if OSPF displayed mssgs, OSPF would appear in this section.

severity: a number that indicates severity of logged event. Some mssgs are just informational, letting us know something has happened while some may be serious & need immediate action. There are 8 severity levels & we need to know all for exam :)

MNEMONIC: short code for mssgs that indicate what happened.

description: this is the detailed information about the event being reported, about what actually happened.

* Syslog format.

seq: time stamp: %. facility - severity - MNEMONIC: description

Syslog Severity Levels

Level	Keyword	Description
0	Emergency	System is unusable.
1	Alert	Action must be taken immediately
2	Critical	Critical Conditions
3	Error	Error conditions
4	Warning	Warning conditions
5	Notice	Normal but significant condition (Notification)
6	Informational	Informational mssgs
7	Debugging	Debug-level mssgs

MNEMONIC - EAC EWO NID
Notice is also displayed as Notification sometimes

- Every Awesome Cisco Engineer Will Need Ice-Cream Daily

Syslog Logging Locations

- Console line:
  Syslog mssgs will be displayed in the CLI when connected to device via console port. By default, all messages (level 0 - level 7) are displayed.

  In packet tracer, when we click device & go to CLI Tab, It operates as if we're connected via console port of device.

- VTY lines:
  Syslog messages will be displayed in the CLI when connected to the device via Telnet/SSH. ~~These~~ This is disabled by default. Logging to VTY lines is disabled by default, so syslog messages will not be displayed if we're connecting device via telnet or SSH.

- Buffer (of device)
  - Syslog mssgs will be saved in to RAM. By default, all mssgs (level-0 - level 7) are displayed.
    ↳ You can view the mssgs with 'show logging' command.

- External server.
  - We can configure the device to send Syslog messages to an external server. This is very useful in large as well as small network.
  - Having ~~syslog mssgs~~ a central server for Syslog mssgs makes network management easier & makes it easier to compare logs of multiple devices.

NOB: Syslog servers will listen for mssgs on UDP Port 514.

Syslog Configuration

i) configure logging to console line. This is enabled by default. Then too,
R1(config)# logging console 6
cmd is logging console level, either level or keyword can be used. This (wl 6) will enable logging for the informational ~~security~~ severity & higher. By default console logs all mssgs.

Scanned with CamScanner

II) configure logging to the vty lines
R1(config)# logging monitor informational.
* We used keyword instead of level. It enables logging for level 0,1,2,3,4,5,6. ~~By default console~~

III) logging to buffer.
R1(config)# logging buffered 51926.
and is logging buffered [size] level.
* same points as above about the level.
* buffer size in bytes.
• Remember no to set buffer size to large.

IV) Logging to an external server.

R1(config)# logging 192.168.1.100
R1(config)# logging host 192.168.1.100
R1(config)# logging trap debugging
• logging server-ip
• logging host server-ip.
* these commands are same.
To configure level of mssgs sent to external server, we use command 'logging trap level'

terminal monitor.

• Even if 'logging monitor level' is enabled, by default Syslog messages will not be displayed when connected via Telnet or SSH.
We need another command to.

- For the mssg to be displayed, we must use following command:
  R1 # terminal monitor.

- This command must be used ~~every~~ every time you connect to the device via Telnet or SSH.

logging synchronous (not essential but very convenient).

- By default logging messages displayed in the CLI while we're in the middle of typing a command will result in something like this:
  R1(config) # exit
  R1 # show ip int
  *May 26 09:30:41.067: %SYS-
  console interface brief

- To prevent this, we should use the "logging synchronous" on the appropriate line. Eg.
  R1(config) # line console 0
  R1(config-line) # logging synchronous

- This will cause a new line to be printed if your typing is interrupted by a mssg.

service timestamps / service sequence number

R1(config) # service timestamps log ?
  datetime Timestamp with date & time
  uptime " System uptime.
  <cr>

R1(config)# service timestamps log datetime
R1(config)# service sequence-numbers
R1(config)# exit.

{service sequence-numbers enable seq nos}

Syslog Command Summary.

i) R1(config)# logging console severity.
ii) R1(config)# logging monitor severity.
iii) R1(config)# logging buffered [size] severity.
iv) R1(config)# logging server-ip.
v) R1(config)# logging host server-ip.
vi) R1(config)# logging trap severity.
vii) R1# terminal monitor
viii) R1(config-line)# logging synchronous
ix) R1(config)# service timestamps log. [datetime | uptime]
x) R1(config)# service sequence numbers